

Ali Azhar Dewali

Biomedical Engineering Student — Brain-Computer Interfaces & Embedded Systems

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EDUCATION

B.Sc. Biomedical Engineering — Double Degree · University of Duhok (UoD) & TH Mittelhessen (THM), Germany

4th-year student · degree taught fully in English · expected graduation 2027. Six-month biomedical engineering internship in Germany planned for 2027 as part of the program.

Secondary education: Werar High School, 2023.

EXPERIENCE

Biomedical Engineering Intern — Dr. Ghazi Hospital, Duhok · Jul–Sep 2025

- Operation, repair, calibration and installation of clinical medical devices; supervised by Eng. Mohammed Mizori.
- Maintenance and inspection of high-capacity oxygen and CO₂ generation plants; troubleshooting of hospital-wide medical vacuum systems used in critical care.
- Recognized for professional conduct and commitment to hospital safety standards.

TECHNICAL PROJECTS

- **EEG-Based Motor Speed Control System Using Brain-Computer Interface (flagship)**. NeuroSky EEG headset → Python engine measuring focus in real time → ESP32-driven fan levitating a ping-pong ball. Full pipeline: acquisition, signal parsing, motor control. Presented at the 13th Engineering Design Day, UoD (25 May 2026). Code: github.com/aliazhardewali-pixel/EEG-Levitation-BCI
- **ECG Monitor Design**. Functional simplified ECG using three bio-potential electrodes to capture and display the heart signal.
- **Microcontroller Motor Control**. DC motor speed control on Arduino Uno/Nano and ESP32 with closed-loop PID and live LCD feedback.
- **Patient Intake Sheet Tool**. Browser tool for physicians — fill in patient data, print a clean one-page intake sheet.
- **C++ Calculator**. Console calculator with input validation and clean control flow.

SKILLS

Programming: C++, Python · **Engineering software:** MATLAB, Dymola, PlatformIO, Arduino IDE, Visual Studio

Data & productivity: Power BI, MS Office · **Hardware:** ESP32, Arduino, circuit design

CERTIFICATIONS & TRAINING

- Certificate of Participation — 13th Engineering Design Day, University of Duhok (25 May 2026).
- Leadership Training (22h) — Kurdistan Students Development Organization, Aug–Sep 2023.
- Rules of Success Course (7h) — Infinity Organization, 2020 · English Language Training — Da Vinci Institute, 2020.
- Python Programming Course (13h) — "Data with Baraa", YouTube (self-study).

LANGUAGES

Kurdish (Badini) — native · Arabic — fluent · English — fluent (degree taught in English)